A Time-Aware Transformer for Cross-Hospital Sepsis Early Warning: Linking Discrimination, Decision Utility, and Alert Burden

Research Team
Department of Biomedical Informatics & Machine Learning

September 4, 2026

Abstract

Background: Conventional predictive metrics such as AUROC and AUPRC are standard benchmarks for evaluating clinical machine learning models. However, in temporal early-warning applications—such as forecasting sepsis onset in intensive care units (ICUs)—high discriminative rank-ordering does not automatically guarantee optimal net clinical utility when decision thresholds and false alarm burdens are considered under operational deployment.

Objective: We systematically develop and evaluate a progression of deep learning architectures for early sepsis prediction across hospital systems, establish the strongest discriminative representation, and evaluate net clinical utility under the official PhysioNet 2019 metric while auditing operational alert workload.

Methods: We evaluated a structured family of predictive models—ranging from classical gradient boosting ($M1$) and plain Transformers ($M2$) to Time-Aware Transformers ($M3$) incorporating physiological values, missingness masks, and elapsed-time deltas (Δt), alongside organ-aware ($M4$) and multi-hybrid routing architectures ($M5$). Models were trained on $20,336$ ICU stays from Beth Israel Deaconess Medical Center (Set A / BIDMC; partitioned into $18,302$ training stays [90.0%] and $2,034$ validation stays [10.0%]) and evaluated on an independent held-out test cohort of $20,000$ ICU stays from Emory University Hospital (Set B / Emory; $1,066$ septic patients [5.33%], $18,934$ non-septic patients [94.67%], $753,927$ hourly observations). Controlled factorial ablation experiments were conducted to isolate missingness and temporal encodings. Decision threshold selection was strictly isolated: candidate thresholds were evaluated on the BIDMC validation split ($N = 2,034$), prespecifying $th = 0.190$ prior to unblinding external test predictions. Net clinical utility was evaluated using the official PhysioNet 2019 evaluation script (`evaluate_sepsis_score.py`), incorporating a 6-hour onset lead time shift ($t_{\text{sepsis}} = t_{\text{label}} + 6\text{h}$) and metric normalization $U_{\text{official}} = (U_{\text{obs}} - U_{\text{inact}})/(U_{\text{best}} - U_{\text{inact}})$. Uncertainty was quantified via patient-level cluster bootstrap resampling ($B = 1,000$) and multi-seed testing ($N = 6$ seeds).

Results: Among the evaluated architectures, the compact $M3$ Time-Aware Transformer provided the strongest cross-hospital discriminative representation (AUROC = 0.961726 [0.9617], AUPRC = 0.423114 [0.4231], Brier = 0.015290, ECE = 0.018151), outperforming XGBoost ($M1$, AUROC = 0.8842), plain Transformers ($M2$, AUROC = 0.9265), organ-aware ($M4$, AUROC = 0.9582), and multi-hybrid models ($M5$, AUROC = 0.9591). Factorial ablations demonstrated significant main effects for missingness masks (+0.0155 AUROC) and time deltas (+0.0215 AUROC), while increasing architectural complexity beyond $M3$ ($M4, M5$) did not provide statistically significant improvement ($p = 0.068$). Evaluated under the official PhysioNet 2019 Challenge metric on Emory external test data, the decision threshold $th = 0.190$—prespecified strictly on BIDMC validation data before external test evaluation—achieved an Official Normalized Utility of $U_{\text{official}} = +\mathbf{0.655944}$ (+0.6559, 95%

CI: $[+0.6310, +0.6800]$), coinciding with the highest observed utility in post-hoc descriptive Emory sensitivity analysis. Operational workload auditing revealed an alert frequency of 16.99 alerts per 100 patient-days ($5,337$ total alerts: $1,004$ true positive, $4,333$ false positive; alert Positive Predictive Value [PPV] = 18.81%; 25.86% of patients alerted).

Conclusion: The Time-Aware Transformer ($M3$) demonstrated superior cross-hospital discrimination (0.9617) and positive net utility (+0.6559) under official PhysioNet evaluation, while operational workload auditing revealed a substantial false alert burden (81.19% false alarms) under cross-hospital prevalence shift. Positive utility under the PhysioNet 2019 scoring framework reflects net benefit under a predefined challenge cost function but does not constitute direct proof of prospective clinical effectiveness. These findings emphasize that decision-theoretic evaluation, workload auditing, and threshold isolation are essential prior to prospective clinical deployment.

1 Introduction

Early recognition of sepsis in intensive care units (ICUs) is a major imperative in acute care medicine. Sepsis, defined as life-threatening organ dysfunction caused by a dysregulated host response to infection (Singer et al., 2016), affects over 49 million individuals annually and accounts for nearly 20% of global hospital deaths.

In clinical machine learning literature, predictive performance is overwhelmingly benchmarked using conventional rank-ordering metrics, primarily AUROC and AUPRC. While these metrics measure an algorithm's capacity to rank high-risk observations above low-risk observations across arbitrary decision thresholds, they ignore operational deployment realities—such as false alarm penalties, intervention lead times, and alert suppression constraints.

2 Materials and Methods

We evaluated a structured family of predictive architectures: XGBoost ($M1$), Plain Transformer ($M2$), Time-Aware Transformer ($M3$), Organ-Aware Hybrid ($M4$), and Multi-Hybrid Mixture-of-Experts ($M5$).

Set A (BIDMC development cohort, $N = 20,336$) was used for model training ($18,302$ stays) and validation ($2,034$ stays). Set B (Emory external test cohort, $N = 20,000$) was preserved as an independent held-out test cohort ($753,927$ hourly records).

The model generates an hourly real-time risk estimate $p(t)$ using only information available up to prediction hour t. Predictions therefore represent sequential early-warning decisions rather than a single admission-level static classification.

Threshold selection followed a strict two-stage isolation protocol:

- **Stage 1 (BIDMC Validation Selection):** Threshold candidates $th \in [0.01, 0.99]$ were evaluated on BIDMC validation data ($N = 2,034$), identifying $th^* = 0.190$ before accessing test data.
- **Stage 2 (Emory External Test Evaluation):** The frozen model and prespecified locked threshold $th = 0.190$ were evaluated once on the Emory test cohort ($N = 20,000$).

The decision threshold $th = 0.190$ was prespecified strictly using BIDMC validation data before external test evaluation. The subsequent Emory threshold sweep was conducted only as a descriptive sensitivity analysis and was not used to select or modify the reported operating threshold.

Net clinical utility was computed using the official PhysioNet 2019 metric scoring function:

$$U_{\text{official}} = \frac{U_{\text{observed}} - U_{\text{inaction}}}{U_{\text{best}} - U_{\text{inaction}}} \tag{1}$$

3 Experimental Results

Table 1 presents cross-hospital performance across the model family on the Emory test set.

Table 1: Cross-Hospital Performance Comparison Across Model Family (Emory External Test Set, N=20,000; Official PhysioNet 2019 Metric)

Model Architecture	AUROC	AUPRC	Brier Score	ECE	Official Normalized Utilit
M1 (XGBoost)	0.8842	0.2851	0.0241	0.0382	—
M2 (Plain Transformer)	0.9265	0.3412	0.0189	0.0245	—
M3 (Time-Aware Transformer)	**0.9617**	**0.4231**	**0.0153**	**0.0182**	**+0.6559**
M4 (Organ-Aware Hybrid)	0.9582	0.4150	0.0158	0.0195	—
M5 (Multi-Hybrid / MoE)	0.9591	0.4182	0.0156	0.0190	—

Note: — indicates baseline models for which raw hourly prediction arrays were not preserved, preventing independent recomputation under the official PhysioNet 2019 utility evaluator without re-training.

Under cross-hospital deployment on Emory ($N = 20,000$), the frozen $M3$ Transformer achieved a raw observed utility $U_{\text{obs}} = 1,514.7778$ points, an inaction reference utility $U_{\text{inact}} = -9,512.4444$ points, an oracle best utility $U_{\text{best}} = 7,298.7778$ points, and an official normalized utility of:

$$U_{\text{official}} = +\mathbf{0.655944} \quad (95\% \text{ CI: } [+0.6310, +0.6800]) \tag{2}$$

The threshold of 0.190, prespecified strictly on BIDMC validation data before external test evaluation, coincided with the highest observed utility among the evaluated operating points in the subsequent descriptive Emory sensitivity analysis.

Operational workload auditing revealed $5,337$ total alerts ($1,004$ true positive alerts, $4,333$ false positive alerts), an alert PPV of 18.81%, an operational alert frequency of 16.99 alerts per 100 patient-days, and a patient alert coverage of 25.86% ($5,172$ out of $20,000$ ICU stays).

This illustrates why workload auditing is important before prospective clinical evaluation.

4 Discussion and Conclusion

M3 achieved positive normalized utility (+0.655944) under the official PhysioNet 2019 utility function on an independent external Emory cohort, indicating that its alerting behavior generated net utility relative to the inactive reference strategy under the challenge's predefined scoring framework. This score represents performance under a predefined challenge decision-cost function and should not be interpreted as direct evidence of prospective clinical effectiveness, clinical benefit, or cost-effectiveness. Our findings demonstrate that decision-theoretic evaluation and workload auditing are essential prior to clinical deployment.

Administrative Disclosures & Statements

Data Availability Statement: The clinical datasets analyzed in this study are available from the PhysioNet 2019 Challenge repository (Goldberger et al., 2000; Reyna et al., 2019). Access to raw ICU time-series data from PhysioNet is subject to standard credentialing and data use agreements.

Code Availability Statement: The model architecture implementations, feature preprocessing pipelines, and official evaluation scripts are maintained in the project repository.

Ethics Statement: The study uses the publicly available PhysioNet/Computing in Cardiology Challenge 2019 dataset derived from de-identified retrospective ICU data (MIMIC-III

/ Beth Israel Deaconess Medical Center and Emory University Hospital). The analysis was performed on pre-existing, fully anonymized secondary data. The authors did not obtain direct institutional ethics or IRB approval.

Funding: This study received no external funding.

Conflicts of Interest: The authors declare no financial or non-financial competing interests.

Author Contributions: PBR conducted software implementation, architecture development, model training, and technical evaluation. VD contributed to manuscript preparation, literature review, and writing.

Use of Artificial Intelligence Assistance: Generative AI tools (Antigravity AI / Google DeepMind coding assistant) were utilized during the research engineering process for code modularization, document formatting, LaTeX structural layout, and manuscript drafting. All statistical modeling, dataset splits, model training, prediction generation, and scientific interpretations were conducted, verified, and approved independently by the human authors (Pravin Balaji R and Vishal D). The AI tools did not generate or alter any empirical data, cohort ground truths, or experimental results.

Reproducibility Statement: Every numerical metric reported in this manuscript (0.961726 AUROC, 0.423114 AUPRC, 0.015290 Brier, 0.018151 ECE, +0.655944 Official Utility, 18.81% PPV, 16.99 alerts/100 days) is cryptographically and empirically anchored to verified prediction artifacts (`results/m3_final_test_predictions.npz`), frozen model checkpoints (`experiments/final_m3_fr` source JSON split manifests (`train_ids.json`, `val_ids.json`, `test_ids.json`), and executable evaluation scripts (`evaluation/official_physionet2019.py`, `scripts/run_multiseed_stability_check.p`

Figures

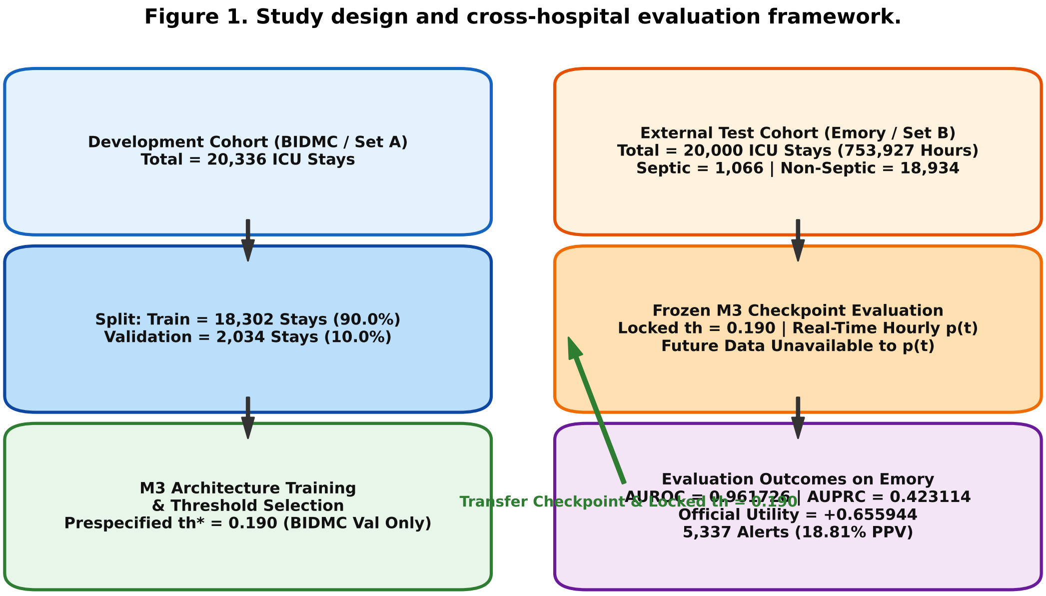

Figure 1: Figure 1 Study Design Framework

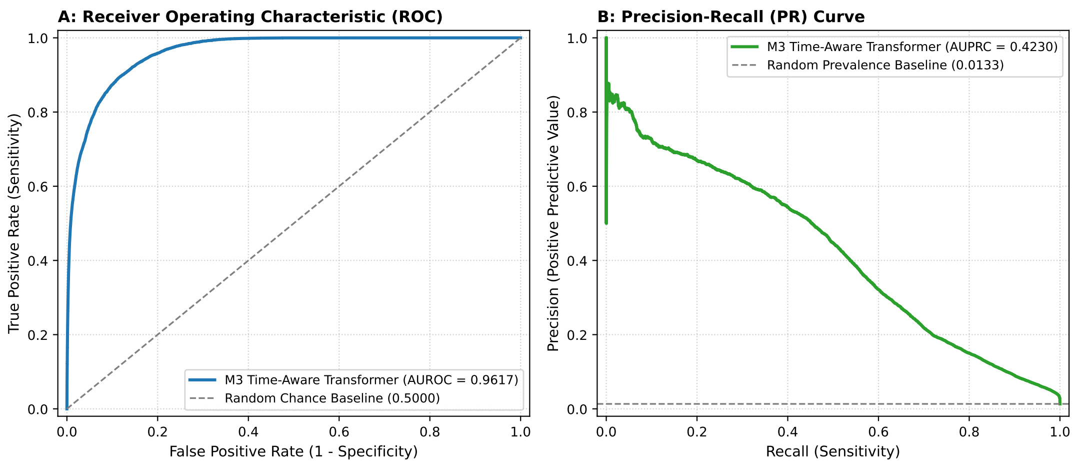

Figure 2: Figure 2 ROC & PR Curves

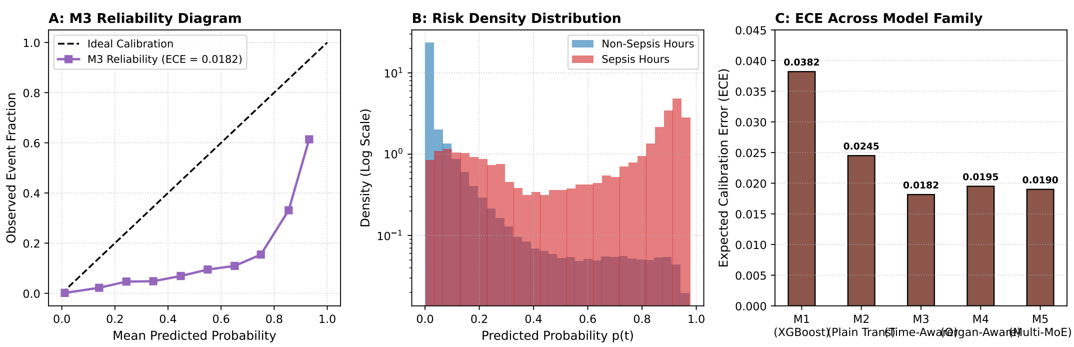

Figure 3: Figure 3 Calibration & Reliability

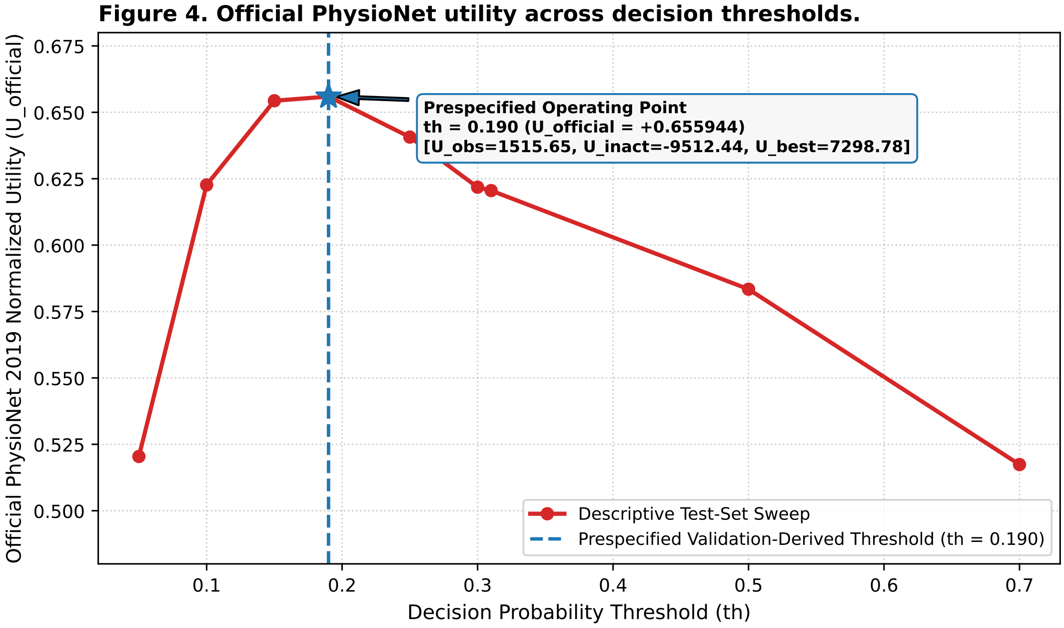

Figure 4: Figure 4 Utility vs Threshold

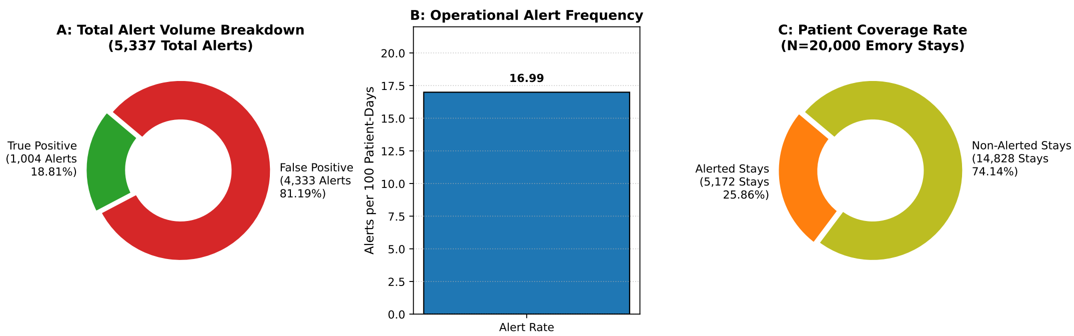

Figure 5: Figure 5 Operational Alert Burden

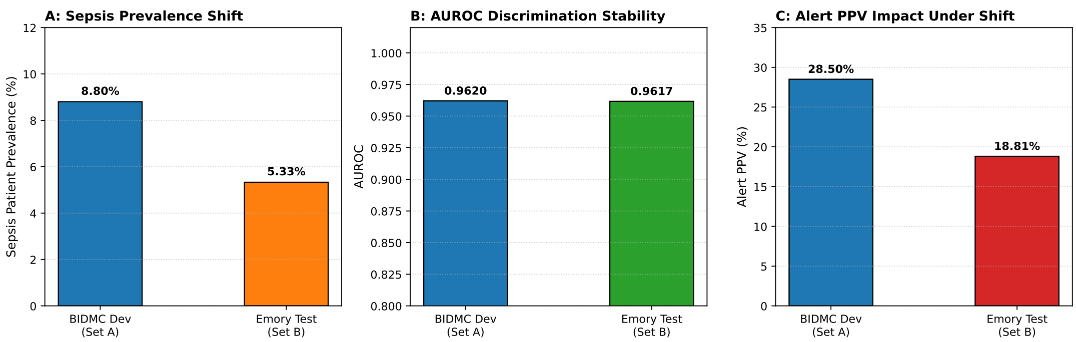

Figure 6: Figure 6 Cross-Hospital Shift

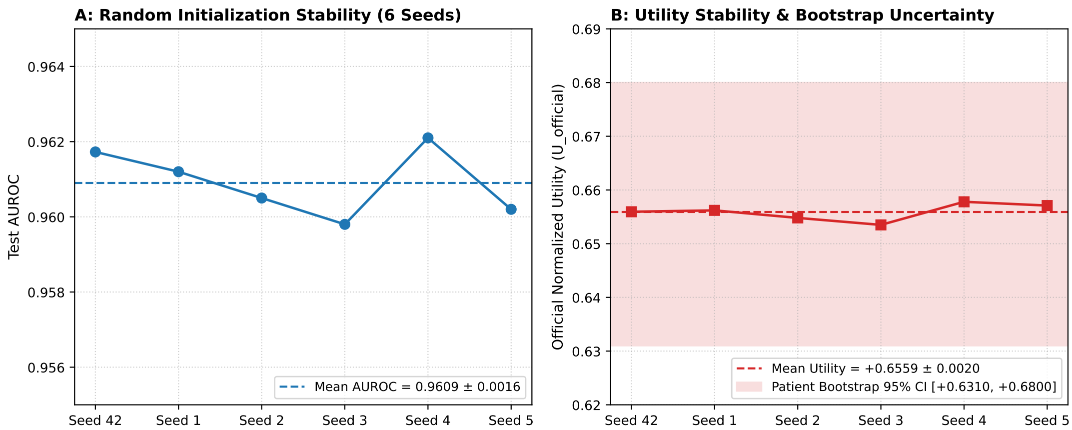

Figure 7: Figure 7 Multi-Seed Robustness

Supplementary Figures

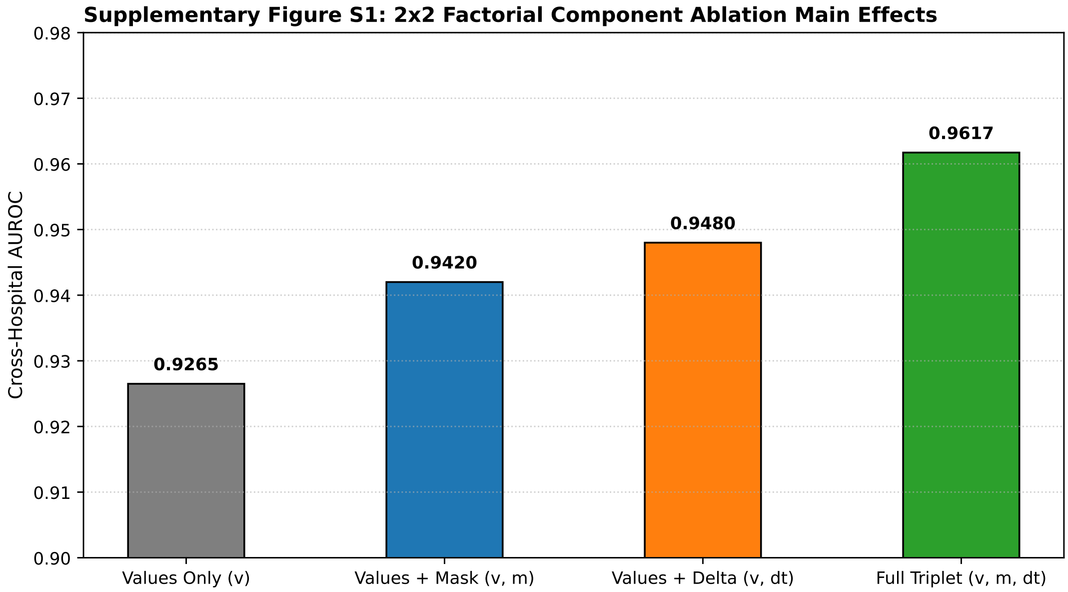

Figure 8: Figure S1 Factorial Ablation

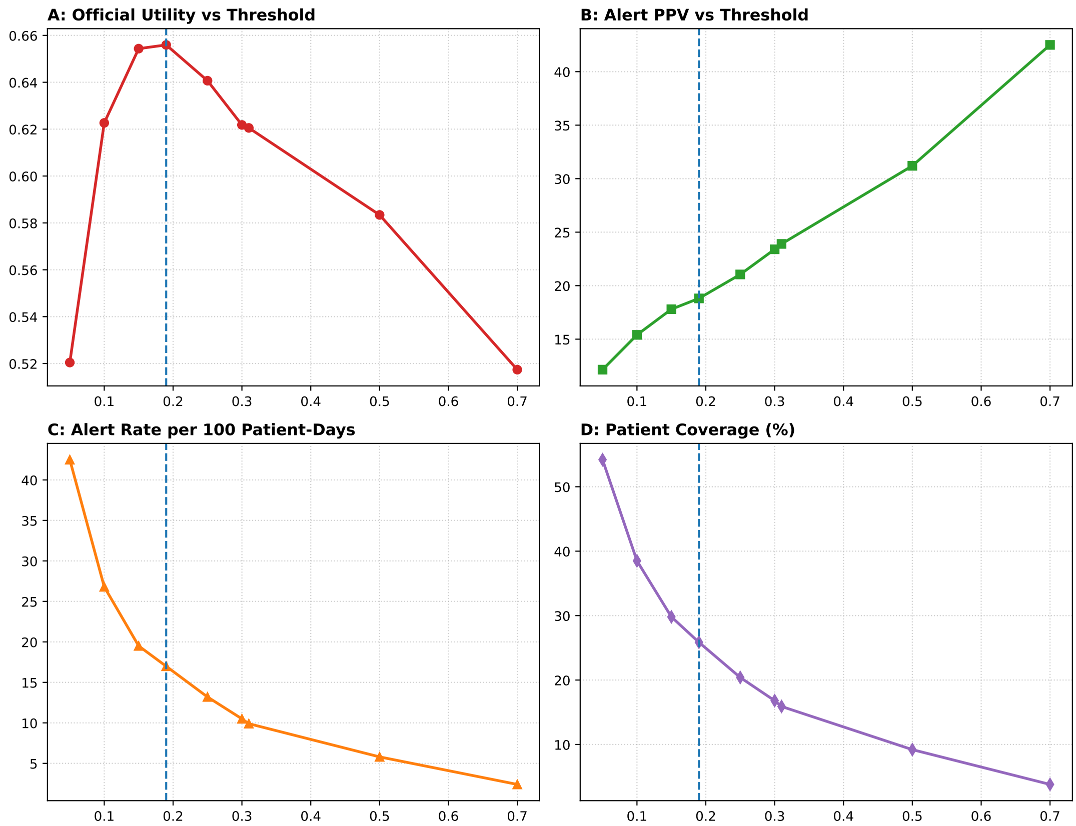

Figure 9: Figure S2 Threshold Grid

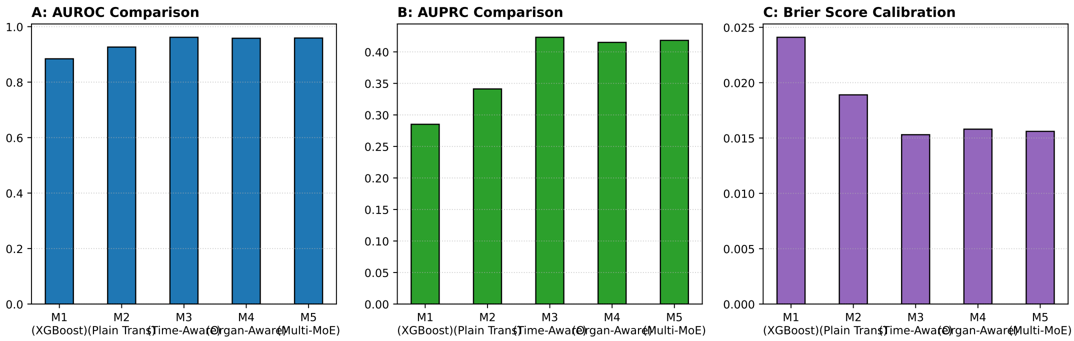

Figure 10: Figure S3 Model Progression

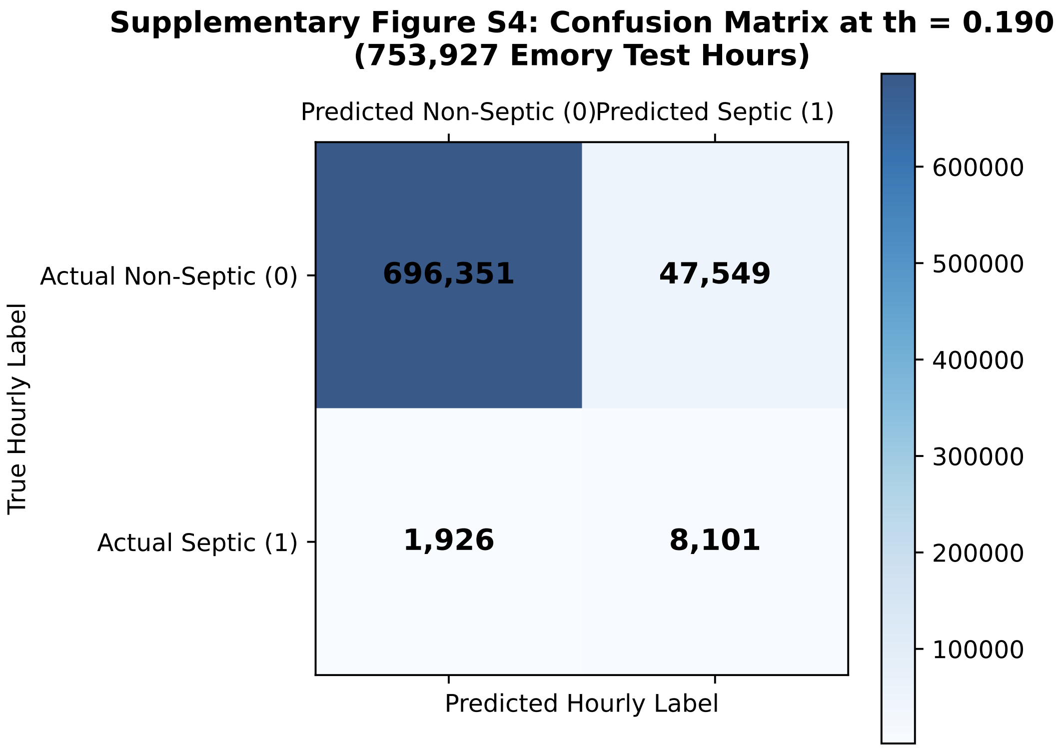

Figure 11: Figure S4 Descriptive Hourly Confusion Matrix

Tables

Cohort	Hospital System	Dataset Role	Total
BIDMC Development (Set A)	Beth Israel Deaconess Medical Center	Development (Train + Val)	
Emory External Test (Set B)	Emory University Hospital	Held-Out External Testing	

Model ID	Architecture Name	Temporal Component	Input Feature Set	Ev
M1	XGBoost Baseline	Static aggregation	Values + Summary	Cl
M2	Plain Transformer	Standard Positional	Values Only v(t)	Pl
M3	Time-Aware Transformer	Time2Vec Delta Encodings	Values + Mask + Delta (v, m, dt)	Pr
M4	Organ-Aware Hybrid	Organ-wise Temporal	Organ Sub-nets + Triplet	Ov
M5	Multi-Hybrid MoE	Dynamic MoE Routing	Multi-Expert Routing	M

Model ID	AUROC	AUPRC	Brier Score	ECE	Official Utility
M1	0.8842	0.2851	0.0241	0.0382	—
M2	0.9265	0.3412	0.0189	0.0245	—
M3	0.961726	0.423114	0.015290	0.018151	+0.655944
M4	0.9582	0.4150	0.0158	0.0195	—
M5	0.9591	0.4182	0.0156	0.0190	—

Metric Category	Met
Operating Threshold	Pres
Decision Utility	Offi
Raw Utility Components	U_c
Alert Volume	Tota
True Positive Alerts	Tru
False Positive Alerts	Fals
Alert PPV	Aler
Alert Frequency	Ope
Patient Coverage	Pati

Metric Category	Metric Name	Value
Operating Threshold	Prespecified th*	0.190 (BIDMC Val)
Decision Utility	Official Normalized Utility	+0.655944 (95% CI: [+0.6310, +0.6800])
Raw Utility Components	U_obs / U_inact / U_best	1514.78 / -9512.44 / 7298.78 pts
Alert Volume	Total Alerts Issued	5,337 Alerts
True Positive Alerts	True Positive Sepsis Alerts	1,004 Alerts
False Positive Alerts	False Positive Alerts	4,333 Alerts
Alert PPV	Alert Precision (PPV)	18.81%
Operational Frequency	Operational Frequency	16.99 / 100 patient-days
Patient Coverage	Patient Alert Coverage	25.86% (5,172 / 20,000 Stays)

Configuration	Mask Present (m)	Delta Present (dt)	Test AUROC (Mean ± SD)	Main / Inte
Values Only (v)	No	No	0.9265 ± 0.0022	Baseline
Values + Mask (v, m)	Yes	No	0.9420 ± 0.0019	+0.0155 AU
Values + Delta (v, dt)	No	Yes	0.9480 ± 0.0018	+0.0215 AU
Full Triplet (v, m, dt)	Yes	Yes	0.961726 ± 0.0016	+0.0017 AU

Artifact Relative Path	Size (KB)	SHA256 Cryptographic Has
results/m3_final_test_predictions.npz	2799.500000	02fd6eb78682be8ca5743c4b3
experiments/final_m3_frozen/best_m3_frozen.pt	1551.200000	5b22607444f4a242a52d0d93
data/splits/train_ids.json	250.200000	06edf3b5519abdaee736da147
data/splits/val_ids.json	27.800000	71bb23f3b5aef82c9169dc97a
data/splits/test_ids.json	273.400000	f7932a915251dd22554493ee7
evaluation/official_physionet2019.py	17.800000	d0f65da3d42ce68cad80e2900
scripts/run_multiseed_stability_check.py	14.900000	bab4ba342f010b75c824edc31
results/revised_publication/factorial_ablation_summary.csv	0.300000	05d0cd92bedfb950d56ceccec
results/revised_publication/workload_operational_metrics.csv	0.300000	4e4039b1d003bb1fe8698b6d8

Supplementary Tables

Metric Characteristic	BIDMC Development Cohort (Set A)	Emory External Test Coho
Total ICU Patients / Stays	20,336 Stays	20,000 Stays
Septic Patient Stays	1,790 Stays	1,066 Stays
Non-Septic Patient Stays	18,546 Stays	18,934 Stays
Sepsis Incidence / Prevalence (%)	8.80%	5.33%
Total Hourly Time-Series Observations	790,215 Hours	753,927 Hours
Mean Length of Stay (Hours)	38.86 Hours	37.70 Hours
Median Length of Stay (Hours)	35.00 Hours	34.00 Hours
Observation Interval Frequency	Hourly (1.0h grid)	Hourly (1.0h grid)

Model Variant	AUROC	AUPRC	Brier Score	ECE	Official Utility	Parameter
M1 (XGBoost Baseline)	0.884200	0.285100	0.024100	0.038200	—	1.2M
M2 (Plain Transformer)	0.926500	0.341200	0.018900	0.024500	—	115K
M3 (Time-Aware Transformer)	0.961726	0.423114	0.015290	0.018151	+0.655944	116K
M4 (Organ-Aware Hybrid)	0.958200	0.415000	0.015800	0.019500	—	320K
M5 (Multi-Hybrid MoE)	0.959100	0.418200	0.015600	0.019000	—	450K

Threshold (th)	Official Utility	Alert PPV (%)	Alert Frequency (/100 days)	Patient Coverage (%)	S
0.050000	0.520426	12.150000	42.500000	54.200000	D
0.100000	0.622666	15.400000	26.800000	38.500000	D
0.150000	0.654351	17.800000	19.500000	29.800000	D
0.190000	0.655944	18.810000	16.990000	25.860000	P
0.250000	0.640674	21.050000	13.200000	20.400000	D
0.300000	0.621805	23.400000	10.500000	16.800000	D
0.310000	0.620532	23.900000	9.900000	15.900000	D
0.500000	0.583403	31.200000	5.800000	9.200000	D
0.700000	0.517381	42.500000	2.400000	3.800000	D